

SET 45 WAVE-PARTICLE DUALITY OF ELECTROMAGNETIC RADIATION

Topic	Problems
Photoelectric effect	1-7
Photons and photon energy	8-11
Relation between wavelength (wave property) and energy (particle property)	12-13
Photon momentum	14-16
Compton effect (without considering the relativistic increase in mass with speed)	17-18
Compton effect (considering the relativistic increase in mass with speed)	19-20
Pressure exerted by electromagnetic radiation	21-24
Pair production	25-26
Interpretation of wave-particle duality, and predominance of wave or particle properties	27-30

Where necessary in this set, use
 speed of electromagnetic radiation in a vacuum or air,
 $c = 3.00 \times 10^8 \text{ m s}^{-1}$
 Planck's constant,
 $h = 6.63 \times 10^{-34} \text{ J s}$
 $= 4.14 \times 10^{-15} \text{ eV s}$
 $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
 $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
 mass of an electron $= 9.11 \times 10^{-31} \text{ kg}$

1 An experiment is carried out with a photoelectric cell using the circuit shown in Figure 45.1. By varying the intensity and frequency of the light used, and by varying the voltage and polarity of the d.c. power supply, the graphs shown in Figure 45.2 were obtained.

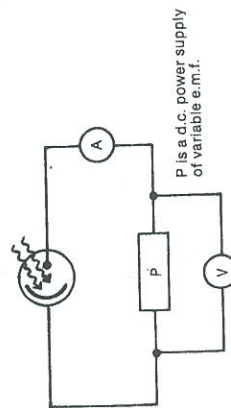


Fig. 45.1

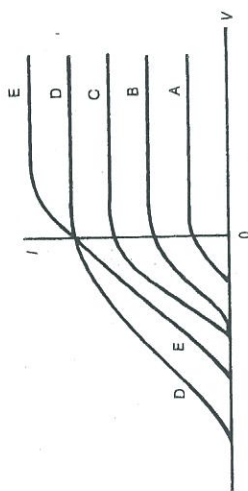


Fig. 45.2

- (a) Which two of the five curves (A, B, C, D, E) in Figure 45.2 correspond to incident light of the same frequency but of different intensities?
 (b) Which of the curves corresponds to the highest frequency of light used?
 (c) Which of the curves corresponds to the situation where the highest energy electrons are ejected from the photoelectric surface?
- 2 The maximum kinetic energy of electrons ejected from a particular metal surface by monochromatic light of frequency ν is measured for several different values of ν . The line of best fit to the experimental points is shown in Figure 45.3.

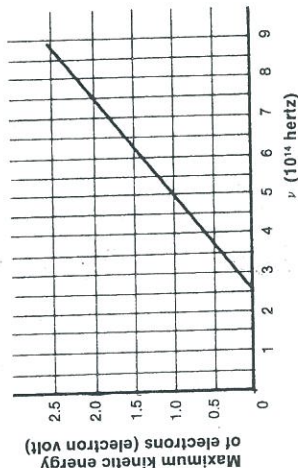


Fig. 45.3

- (a) What is the value of Planck's constant (in eV s) as calculated from the graph?
 (b) What is the minimum energy (in eV) required to extract an electron from this metal surface?
 (c) If deep violet light of frequency $7.5 \times 10^{15} \text{ Hz}$ is used, what will be
 (i) the maximum kinetic energy (in eV) of the ejected photoelectrons?
 (ii) the minimum stopping potential difference that would have to be applied across the photocell to prevent the photoelectrons from reaching the collector electrode?

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3 An experiment on the photoelectric effect was carried out, in which the maximum kinetic energy of electrons emitted from the surface of a particular metal was measured and then plotted against the frequency of the incident electromagnetic radiation. In Figure 45.4 the graph of E_{max} versus ν was obtained using electromagnetic radiation of a fixed intensity I_0 , and shows the line of best fit to the experimental points.

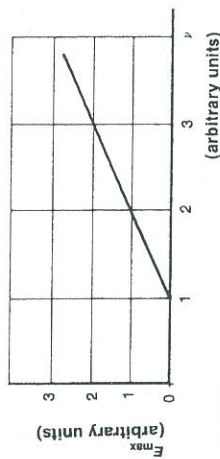


Fig. 45.4

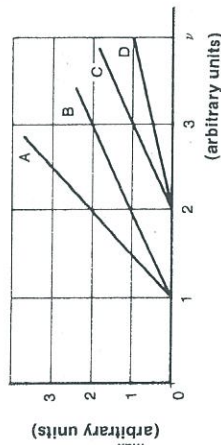


Fig. 45.5

- (a) Which of the graphs A–D drawn to the same scale in Figure 45.5 would be obtained for the same metal, using electromagnetic radiation of intensity $2I_0$?
 (b) Which of the graphs A–D in Figure 45.5 would be obtained using intensity I_0 for another metal for which the minimum energy required to remove an electron is twice that for Figure 45.4?
- 4 Electrons are ejected from a silver surface by radiation of wavelength 253.6 nm. The cut-off potential difference required to stop the photoelectrons is 0.12 V. Determine (in eV)
 (a) the maximum kinetic energy of the ejected electrons;
 (b) the work function of the silver surface.
- 5 The threshold frequency for a rubidium surface is $5.0 \times 10^{14} \text{ Hz}$. Find
 (a) the work function for the metal surface in eV;
 (b) the maximum speed of electrons ejected by light of frequency $8.0 \times 10^{14} \text{ Hz}$.
- 6 Radiation of wavelength 375 nm is incident upon a surface which has a work function of 2.0 eV. Calculate

- (a) the energy (in eV) of each of the incident photons;
 (b) the maximum kinetic energy (in eV) of the ejected electrons;
 (c) the maximum speed of the electrons.

7 The threshold frequency for a particular material is $3.0 \times 10^{14} \text{ Hz}$. If blue light of frequency $7.0 \times 10^{14} \text{ Hz}$ falls on this surface, what is

- (a) the energy (in joule) of the incident photons?
 (b) the work function (in joule) of the material?
 (c) the maximum kinetic energy of the ejected electrons?
 (d) the maximum speed of the ejected electrons?

8 What is the energy (in joule) of a photon of yellow light of frequency $5.1 \times 10^{14} \text{ Hz}$ from a sodium vapour lamp?

9 Radio photons from station 2CA in Canberra have an energy of $6.95 \times 10^{-28} \text{ J}$. On what radio frequency does 2CA transmit?

10 A typical microwave apparatus used in schools uses a 32 mm wavelength.

- (a) What is the frequency of the microwave photons?
 (b) What is the energy of one of the microwave photons?

11 Visible light ranges in wavelength from about 400 nm in the deep violet to about 750 nm for red light. Estimate as orders of magnitude

- (a) the energy of a photon coming from a 100 W household light globe;
 (b) the number of photons leaving the filament of the globe in 1 s.

12 Like many other questions in this set this problem relates *particle* and *wave* properties of electromagnetic radiation.

- (a) Write down an expression for photon energy E (a particle property) in terms of radiation frequency ν (a wave property).
 (b) Deduce an expression for photon energy E (a particle property) in terms of radiation wavelength λ (a wave property).
 (c) Your answer to (b) should be in the form $E = k/\lambda$, where k is a constant. If SI units are used, i.e. E is in joule and λ is in metre, what is the value of k ?
 (d) Often it is useful to be able to relate photon energy in electron volt to radiation wavelength, in nanometre. Deduce the relationship for E (in eV) in terms of λ (in nm), including the numerical value of the constant. This relation is possibly worth memorising.

13 (a) What are the approximate maximum and minimum energies of the photons that enable you to see? Give your answers in eV. The relation deduced in Problem 12 (d) should prove useful. The wavelength of visible light ranges from about 400 to 750 nm.

- (b) What is the wavelength (in nm) corresponding to an x-ray photon of energy 1.24 keV?

ANSWERS

- 18 21 cm
19 74 cm
20 $R/(r-1)$
21 (a) Red light (b) 1.03
22 21.9 cm
23 Focal length is 40 cm; distance from A is 5 cm
24 10.3 cm
25 40 cm
26 15 cm

SET 42 SIMPLE OPTICAL INSTRUMENTS

- 1 (a) Between X and Y, 15 cm from X (Was your answer ambiguous?) (b) 0.50 (c) 40 cm from Y on the side opposite to X (d) 0.83 (e) Right way up
2 (a) 100 cm from P on the side opposite to the bulb (b) 23 cm from Q on the side opposite to the bulb (c) Real and inverted
3 8.8 cm from the concave lens on the side closer to the convex lens; 0.040
4 60 cm behind the lens
5 10 cm
6 (a) 0.24 cm (b) 0.48 cm
7 (a) 15 cm (convex) (b) 15 cm (concave) (c) A concave lens of focal length 30 cm
8 (a) 36 cm (b) 18 cm (c) Concave
9 (a) 3.0 cm (b) 2.7 cm
10 (a) Concave lenses (b) 50 cm
11 1.5 m
12 (a) Concave lens of focal length 1.8 m (b) Convex lens of focal length 50 cm
13 77 cm
14 21 cm
15 20 cm
16 10 cm from the mirror on the side opposite to the lens and 40 cm from the lens on the side opposite to the mirror
17 (a) 1.00 m (b) 10 cm (c) 1.10 m (d) 10
18 (a) 84 cm (b) 7 cm
19 (a) 7.1 cm (b) 1.00 m (c) 14
20 (a) 5.0 (b) 32 cm (c) 1.52 m (d) 5.0
21 (a) 20 (b) 2.50 m (c) The eyepieces are moved towards the objective
22 (a) 6.0 (b) 70 cm (c) 98 cm
23 (a) 2.4 cm (b) 3.6 cm (c) The opera glasses give an upright image whereas the simple telescope would give an inverted image
24 (a) 40 (b) 102.5 cm
25 (a) 9.2×10^{-3} rad; 0.53° (b) 0.15 m (c) 300 (d) 2.7 or 2.8 rad (dependent on method)

254

SET 43 SIMPLE WAVE PROPERTIES OF LIGHT AND OTHER FORMS OF ELECTROMAGNETIC RADIATION

- 26 (a) 5.6 m (b) 6.8×10^6 m
27 (a) 250 nm (b) 4.0 (c) 270 nm (d) 10 (e) 40
28 (a) 3.3 cm (b) 20
29 3.20 cm and 11.2 cm
30 3.0 $\times 10^8$ m s⁻¹
31 4.69 $\times 10^{14}$ Hz (b) 2.13×10^{15} s
32 Red, orange, yellow, green, blue, violet
33 Gamma rays, x-rays, ultraviolet light, visible light, infrared light, microwaves, radio waves
34 (a) Visible light (b) Microwaves (c) X-rays
35 (a) Radio waves (e) All types of electromagnetic radiation (f) All types of electromagnetic radiation
36 (a) 5.20×10^{14} Hz (b) (i) 5.20×10^{14} Hz (ii) 356 nm (iii) 1.85×10^8 m s⁻¹
37 (a) 7.41×10^7 s (b) 222 m (c) 2.0×10^8 m s⁻¹
38 (a) (i) S (ii) X (iii) X (iv) X (b) 1.34
39 Away from us (b) 1.40×10^7 m s⁻¹
40 2.4 nm
41 Approximately 9 Hz
42 Light waves between the threads of the handkerchief (b) Red light (c) Radio waves
43 (a) B (b) W (c) B (d) W (e) W
44 (a) 1.25×10^8 m s⁻¹ (b) 7.20×10^8 m s⁻¹

SET 44 INTERFERENCE AND DIFFRACTION WITH ELECTROMAGNETIC RADIATION

- 1 1 km
2 (a) 3.0 cm (b) 10^{10} Hz
3 0.40 μ m and 0.60 μ m
4 (a) 38.9 cm (b) C D
5 (a) 7q (b) 7/9
6 (a) A bright band centred on G (b) A dark line centred on K (c) 6.0×10^{-4} m
7 (a) 3.60 mm (b) 4.80 mm
8 2.21 mm

- 9 630 nm
10 420 nm
11 (a) $\frac{1}{2}$ (b) 19°
12 (a) 7 (b) 0.25
13 (a) 0.44 mm (b) $\frac{1}{2}$ (c) X is on the third bright band from C
14 (a) 4, including one at P (b) $5 \times 10^3 \lambda$
15 (a) $D = A(r-1)$ (b) 0.12 mm
16 (a) Bright light (b) Bright light (c) Dark line
17 (a) Bright light (e) Dark line
18 (a) (i) λ (ii) $5\lambda/2$ (iii) 3λ (b) 4.2 mm
19 (a) 1.40 μ m (or 1400 nm) (b) 90 mm (c) 2.1×10^{-4} m
20 (a) 36 mm (b) $R/2$
21 36 μ m
22 (a) 1.5×10^{10} m (b) 69 m
23 6 km
24 B C E
25 392 nm
26 Second order
27 600 lines mm⁻¹
28 4
29 393 nm
30 (a) Yes (b) (i) No (ii) No (c) (i) Blue (ii) Black (d) 90 nm (e) $3\lambda/4$
31 (a) (i) Out of phase (ii) In phase (b) Reflected light (c) 590 nm (d) b and d
32 (a) 399 nm (b) 100 nm
33 (a) Dark band (b) 295 nm (c) 92 fringes (d) 27 μ m
34 0.24 mrad, 0.014°
35 44 μ rad
36 (a) (i) $\lambda/4$ (ii) $3\lambda/4$ (iii) $\lambda/2$ (iv) λ (b) $\lambda/2$ (c) 7A (d) $d = m\lambda/2$ (e) $R = (r^2 + d^2)/2d$ or, better still, $R \approx r^2/2d$, since $r^2 \gg d^2$ (f) $R \approx r^2/m\lambda$
37 (a) 698 nm (b) 670 nm
38 391 nm
39 1.42
40 (a) $2d \sin \theta$ (b) $2d \sin \theta = m\lambda$, where $m = 0, 1, 2, \dots$ (c) 22° and 48°
41 $18^\circ, 38^\circ, 66^\circ$
42 0.314 nm
43 74.8 μ m
44 176 μ m

SET 45 WAVE-PARTICLE DUALITY OF ELECTROMAGNETIC RADIATION

- 1 (a) B C (b) D (c) D
2 (a) 4×10^{-15} eV s (b) 1.0 eV (c) (i) 2.0 eV (ii) 2.0 V

ANSWERS

- 3 (a) B (b) C
4 (a) 0.12 eV (b) 4.78 eV
5 (a) 2.1 eV (b) 6.6×10^5 m s⁻¹
6 (a) 3.3 eV (b) 1.2 eV (c) 6.8×10^5 m s⁻¹
7 (a) 4.6×10^{-19} J (b) 2.0×10^{-19} J (c) 2.6×10^{-19} J (d) 7.7×10^5 m s⁻¹
8 3.4×10^{-19} J
9 1.05 MHz
10 (a) 9.4×10^8 Hz (b) 6.2×10^{-24} J
11 (a) 10^{-18} J (b) 10^{20} photons s⁻¹
12 (a) $E = h\nu$ (b) $E = hc/\lambda$ (c) $k = 1.99 \times 10^{-25}$ J m (d) $E = 1.24 \times 10^3/\lambda$
13 (a) Approximately 3.10 to 1.65 eV (b) 1.00 nm
14 1.05×10^{-27} kg m s⁻¹ west
15 3.00×10^{-13} m
16 (a) $p = h\nu/c$ (b) 1.82×10^{-27} kg m s⁻¹ (c) $p = E/c$
17 (b) 6.60×10^{-24} kg m s⁻¹ east, 6.60×10^{-24} kg m s⁻¹ north (c) Answer should be within the range (9.32 to 9.34×10^{-24} kg m s⁻¹ south-east (d) Answer should be within the range (4.76 to 4.80×10^{-17} J (e) 1.98×10^{-15} J (f) 1.93×10^{-15} J (g) 2.5 per cent error (h) 3 pm
18 (a) 0.620 (or 0.622) pm (b) 1.07×10^{-21} kg m s⁻¹ north (c) 5.35×10^{-22} kg m s⁻¹ N 60° W (d) 1.24 pm (e) 0.62 pm
19 (i) Einstein showed that for speeds approaching that of electromagnetic radiation the effective increase in mass with speed was given by this relation (ii) Conservation of energy (iii) Conservation of momentum in the x-direction (iv) Conservation of momentum in the y-direction
20 172 keV
21 (a) $h\nu/c$ east (b) $2h\nu/c$ west (c) $2N/h\nu/c$ west (d) $2N/h\nu/c$ west (e) $2N/h\nu/c$ east (f) $2N/h\nu/c$ (g) $E = h\nu$ (h) $2E/c$ This expression for the pressure would still hold if the average frequency of the polychromatic light equalled the frequency of the monochromatic light (c) 10^6 Pa (d) 10^{10} greater
22 (a) 1.21×10^{-27} kg m s⁻¹ (b) 1.21×10^{-27} kg m s⁻¹ (c) 3.62×10^{-19} J (d) 1.7×10^{20} photons s⁻¹ (e) 1.7×10^{20} photons s⁻¹ (f) 5.5×10^{-4} (g) 9.4×10^{16} photons s⁻¹ (h) 1.1×10^{-10} kg m s⁻¹ (i) 1.1×10^{-10} N (j) 1.1×10^{-10} N (k) 1.8×10^{-9} Pa
23 Approximately 10^{15} W
24 1.02 MeV
25 (a) 0.58 MeV (b) 1.60 MeV
26 B C D F
27 B C D F
28 (a) F (b) D
29 (a) Lower frequency (b) Shorter wavelength
30 B C F

255